

PRATIK KAMATH

☎ +61 433 839 218 ✧ Visa: Subclass 485 (Valid until Oct 2028, full Australian work rights)
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Experience

Multifactor AI Inc

Feb 2026 – Present

Artificial Intelligence Engineer

California, US (Remote)

- Built [Porchflip](#)'s semantic retrieval engine (**FastAPI** on **Cloud Run**): **pgvector** cosine search over **Vertex AI**, feeding a combinatorial optimizer that assembles budget-constrained furniture kits across 6 roles.
- Engineered the **2,000+ SKU** embedding pipeline: a scraper (**curl_cffi** and **patchright**) feeding **Vertex AI** embeddings into **pgvector**, plus keyless **GitHub Actions** CI/CD (**OIDC**/WIF).
- Built a custom **LangGraph tool-calling** scheduling copilot (**10 tools**) that re-validates LLM proposals under row locks before commit, guarded by a circuit breaker and fail-closed admission control.
- Architected the production **FastAPI** backend behind these systems: **64 REST endpoints**, **33 SQLAlchemy** models across 8 domains, **Alembic** migrations, and a **660-test pytest** suite on PostgreSQL.
- Implemented **JWT** auth with single-use refresh-token rotation and a **9-role RBAC** layer with department scoping, plus append-only audit logging across mutating modules for compliance.

Mark Computers

July 2025 – Dec 2025

Software Engineer

Mumbai, India

- Engineered enterprise-grade Fuel Retail Management System automating POS, accounting, inventory, and HR for multi-station petrol pump facilities, deployed on **AWS (EC2, RDS)**; reduced manual logging errors by **93%**.
- Automated 20+ operational workflows reducing reconciliation time by **75%**, and designed multi-shift workforce management for 30+ staff with attendance, payroll, and role-specific permissions unified into a single system.

L&T Realty

May 2021 – Jul 2021

Software Engineering Intern

Mumbai, India

- Trained and tuned Random Forest and Gradient Boosting models with **scikit-learn**, lifting price prediction accuracy from a **74% baseline to 92%** validated through 5-fold cross-validation.
- Built and deployed a full-stack real estate valuation platform using **FastAPI**, **React**, and **PostgreSQL**, automating pricing across 5,000+ listings and sustaining **1,000+ daily API requests** during internal testing.

Technical Skills

AI & ML: LLM agents & tool-calling (OpenAI SDK), RAG & vector search (pgvector), Vertex AI embeddings, scikit-learn
Languages: Python, JavaScript/TypeScript, C, SQL
Backend & Data: FastAPI, REST APIs, SQLAlchemy, Alembic, PostgreSQL, pgvector, Redis, MongoDB
Cloud & MLOps: GCP (Cloud Run, Cloud SQL, Vertex AI), AWS (EC2, RDS), Docker, GitHub Actions CI/CD
Frontend & Tools: React, React Native (Expo), Next.js, pytest, Git, Agile/Scrum

Education

Master of Information Technology, Major in Artificial Intelligence

June 2025

University of New South Wales

Sydney, NSW

- Recipient of the **International Student Award** for **academic merit** and commitment to studies.
- Distinction WAM** in core courses including Advanced Machine Learning and Data Structures & Algorithms.

Bachelor of Computer Engineering

May 2022

NMIMS University

Mumbai, India

- Published research on **financial time series forecasting using neural networks**, a comparative study [\[IEEE\]](#)

Projects

Conference Management Platform [\[GitHub\]](#) | React, React Native, Node.js, Docker, MongoDB, Redis

2025

- Spearheaded 6-member agile team as **Product Owner** through full SDLC (design, code reviews, testing, deployment) to deliver a full-stack platform serving **500+ users** across web and mobile.
- Designed scalable **Node.js** backend with microservices-inspired architecture, integrating **MongoDB**, **Redis** caching, and **JWT authentication**, achieving **sub-200ms API response times**.
- Implemented **WebSocket**-based real-time chat and Q&A forums with push notifications via Expo, orchestrated **Docker** containerization and **CI/CD** deployment workflows.

Certifications

AWS: Cloud Quest in Cloud Practitioner & Generative AI Practitioner

Databricks: Academy Accreditation in AI Agent Fundamentals & Generative AI Fundamentals